

Chemistry, 2016 EC, model exam, Arada

Grade 12 · Chemistry · 2016 E.C.

Time allowed: 120 minutes

Questions: 78

Total marks: 78

Name _____ Grade _____ ID No. _____ Date _____

1. Which model of an atom describes an atom as a positive sphere where electrons are embedded in a sphere? (1)
 A. Ernest Rutherford B. J.J. Thomson
 C. Neil's Bohr D. John Dalton
-
2. An element X has two stable naturally occurring isotopes ($^{16}_8X$ & $^{18}_8X$). If the average atomic mass of element X is 16.2 a. m. u. what is the percentage abundance of $^{16}_8X$? (1)
 A. 90 B. 25
 C. 10 D. 75
-
3. Which of the following property decreases across a period of the periodic table? (1)
 A. Ionization energy B. Atomic size
 C. Effective nuclear charge D. Electron affinity
-
4. Unequal distribution of the shared electrons between two atoms results in the formation of - (1)
 A. Polar covalent bond B. Non-polar covalent bond
 C. Electrovalent bond D. Co-ordinate covalent bond
-
5. Which of the following is non-polar molecule? (1)
 A. NH_3 B. HF
 C. CCl_4 D. SO_2
-
6. According to Boyle's law, doubling the pressure of a sample of gas, what will be the volume of gas at constant temperature? (1)
 A. Doubled B. Tripled
 C. Halved D. One third
-
7. The following table shows substance of comparable molecular mass with their dominant intermolecular forces. (1)
 Which of the following is FALSE about the substances?
 [Figure: A table with two columns and three rows of data:
 Column 1 header: Substance
 Column 2 header: Dominant intermolecular force
 Row 1: A | Hydrogen bonding
 Row 2: B | London-dispersion force
 Row 3: C | Dipole –dipole force]
 A. "B" has the weakest intermolecular force attraction
 B. "C" has the highest boiling point than "A"
 C. "A" has the strongest intermolecular force attraction
 D. "C" has the strongest intermolecular than "B"

8. 64g of sulphur dioxide SO_2 is burned in air to produce sulphur trioxide, SO_3 . If the percentage yield of SO_3 is 90%, what is the actual yield of SO_3 ? (1)
- A. 80g
B. 72g
C. 57.6g
D. 160g
9. The following figure shows an energy diagram for a certain chemical reaction (1)
Which of the following is TRUE about the reaction shown in the diagram?
[Figure: An energy diagram with vertical axis labeled 'Energy' and horizontal axis labeled 'Course of reaction'. A horizontal plateau at a higher energy level is labeled 'Reactants'. A downward slope connects 'Reactants' to a lower horizontal plateau labeled 'Products'. A dashed horizontal line extends to the right from the 'Reactants' plateau directly above the 'Products' plateau. Double-headed vertical arrow C extends from the horizontal axis up to the 'Reactants' level. Double-headed vertical arrow B extends from the horizontal axis up to the 'Products' level. Double-headed vertical arrow A extends from the dashed line representing the 'Reactants' level down to the 'Products' level.]
- A. The reaction is endothermic
B. 'C' indicates heat content of product
C. 'A' indicates heat change of the reaction
D. 'B' indicates heat content of reactant
10. If the proportions of reactants in a reaction mixture are non-stoichiometric, which of the following statement is true? (1)
- A. The quantities of the consumed reactant in the reaction maintain stoichiometric proportion
B. The amount of yield is proportional to the amount of excess reactant
C. The excess reactant is always the one with higher number of moles
D. The limiting reactant is the one with less number of moles
11. In an oxidation- reduction reaction, the oxidizing agent is (1)
- A. reduced as it gains electrons
B. oxidized as it loses electrons
C. contains an element that undergoes oxidation
D. brings about reduction but itself oxidized
12. In each of the four beakers shown below, a 2.0cm strip of magnesium ribbon is added to 100mL of aqueous solution of HCl under the given condition shown (1)
In which beaker will the reaction occur at the fastest rate? In the beaker that contains
[Figure: Four cylindrical beakers are shown side by side, each containing an aqueous HCl solution with different conditions labeled inside: first beaker has '0.2M' and '10°C', second beaker has '0.1M' and '30°C', third beaker has '0.1M' and '10°C', and fourth beaker has '0.2M' and '30°C'.]
- A. 0.1M at 30°C
B. 0.2M at 30°C
C. 0.1M at 10°C
D. 0.2M at 10°C
13. The quantity of heat required to convert one mole of a solid into its liquid state is (1)
- A. Molar heat of condensation
B. Molar heat of fusion
C. Molar heat of crystallization
D. Molar heat of sublimation

14. Given the reaction at equilibrium: $X_{2(g)} + 2Y_{2(g)} \rightleftharpoons 2XY_{2(g)} + \text{heat}$
How the equilibrium reaction does yields high amount of product?
A. By increasing both the temperature and pressure
B. By decreasing the temperature and increasing pressure
C. By increasing the temperature and decreasing pressure
D. By decreasing both the temperature and pressure
-
15. Two gases A and B were applied to the two ends of 40 cm long glass tube. The two gases met at a distance of 30 cm from the side where B is applied. Which statement is true about these gases?
A. The density of A is smaller than the density of B
B. The rate of diffusion of A is faster than the rate of diffusion of B
C. The molar mass of A is higher than the molar mass of B
D. The rate of diffusion of B is slower than the rate of diffusion of A
-
16. Which set of compounds represents a homologous series?
A. C_2H_6 , C_3H_8 , C_4H_8
B. C_2H_6 , C_3H_8 , C_4H_{10}
C. C_2H_6 , C_3H_8 , C_5H_{10}
D. C_2H_6 , C_3H_8 , C_4H_6
-
17. Which one of the following is a tertiary alcohol?
A. 3 - Hexanol
B. 1,2,3 - propanetriol
C. 3 - methyl - 2 - pentanol
D. 2 - methyl - 2 - propanol
-
18. What is the correct IUPAC name of the compound shown below?
[Figure: A chemical structural formula showing a central carbon chain: CH_3 bonded to C, which is bonded to a Br above, a CH_2-CH_3 group below, and a CH_2 group to the right. The CH_2 group is bonded to a CH group, which has a CH_2-CH_3 group above it and a CH_3 group to its right.]
A. 2 - bromo - 2,4 - diethyl pentane
B. 3 - bromo - 3,5 - dimethyl heptane
C. 3 - bromo - 5 - ethyl - 3 - methyl hexane
D. 2 - bromo - 2 - ethyl - 4 - methyl hexane
-
19. Which of these statements is FALSE about the identification test of saturated and unsaturated hydrocarbons?
A. Br_2 in CCl_4 is used to identify whether the compound is saturated or not.
B. The consumption of hydrogen gas (H_2) by the compound to be tested can be used as a reagent for the existence of unsaturated hydrocarbon or not
C. The reagent $KMnO_4$ (Alkaline) can be used to test unsaturated hydrocarbon
D. The reagent nitric acid (HNO_3) can be used to test unsaturated hydrocarbon
-
20. Alcohol that yields ketones on oxidation are classified as;
A. Dihydric alcohol
B. Trihydric alcohol
C. Primary alcohol
D. Secondary alcohol
-
21. Which one of the following is not a ternary acid?
A. HCN
B. HF
C. HNO_3
D. H_3PO_2

22. When an acid reacts with an active metal: (1)
- A. hydronium ion concentration increases B. hydrogen gas is produced
C. the metal forms anions D. carbon dioxide gas is produced
23. What is the $[OH^-]$ of a solution with a pH of 4? (1)
- A. 10^{-4} B. 10^4
C. 10^{10} D. 10^{-10}
24. Which one of the following salts is not an acidic salt? (1)
- A. $KHSO_4$ B. Na_2HPO_4
C. $(NH_4)_2SO_4$ D. $Mg(HCO_3)_2$
25. During the electrolysis of molten lead bromide using graphite electrodes, the anode reaction is (1)
- A. $2Br^- \rightarrow Br_2 + 2e^-$ B. $Pb \rightarrow Pb^{+2} + 2e^-$
C. $Br_2 + 2e^- \rightarrow 2Br^-$ D. $Pb^{+2} + 2e^- \rightarrow Pb$
26. As opposed to a galvanic cell, in an electrolytic cell (1)
- A. spontaneous redox reaction generates electricity
B. electrons flow from cathode to anode
C. the anode is positive
D. oxidation takes place at the cathode
27. In the extraction of aluminum from bauxite, bauxite is treated with hot aqueous sodium hydroxide. The reason for this is to (1)
- A. lower the melting point of bauxite B. increase the strength of aluminum
C. displace aluminum from its oxide D. remove impurities from bauxite
28. Which of the following is the purest form of iron? (1)
- A. Cast iron B. wrought iron
C. Pig iron D. limonite
29. Which of the following methods is used to reduce land pollution? (1)
- A. Replacing CFCs by less harmful alternatives
B. Removal of lead compounds from gasoline
C. Reducing CO_2 emission
D. Recycling plastics
30. In the process of converting skin to leather by tanning, the use of slaked lime is to (1)
- A. make sole leather B. remove the hairs
C. kill bacteria and dehydrate D. soften the skin and hide
31. Precision refers to how close a measured value to (1)
- A. the true value B. the other measured values
C. zero D. the calculated value
32. Which of the following represents a tentative explanation of a certain scientific law? (1)
- A. Hypothesis B. Observation
C. Experimentation D. Theory

43. What is the relationship between K_c and K_p for the reaction $3A(g) \rightleftharpoons B(g) + C(g)$ (1)
- A. $K_p = K_c(RT)^1$ B. $K_p = K_c(RT)^{-1}$
 C. $K_p = K_c(RT)^2$ D. $K_p = K_c(RT)^{-2}$
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44. What is the general formula of saturated monocarboxylic acids (1)
- A. $C_nH_{2n+1}O$ B. $C_nH_{2n}O$
 C. $C_nH_{2n+2}O_2$ D. $C_nH_{2n}O_2$
-
45. What is the process that converts vegetable oils to solid fats? (1)
- A. Hydration B. Hydrogenation
 C. Hydrolysis D. Saponification
-
46. What is the parent acid and alcohol to prepare isopropyl pentanoate? (1)
- A. $CH_3 - \underset{\text{textOH}}{\underset{|}{CH}} - CH_3$ & $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH$ B. $CH_3 - \underset{\text{textOH}}{\underset{|}{CH}} - CH_3$ & $CH_3 - CH_2 - CH_2 - CH_2 - \underset{\text{textO}}{\underset{||}{C}} - OH$
 C. $CH_3 - CH_2 - \underset{\text{textO}}{\underset{||}{C}} - OH$ & $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH$ D. $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - \underset{\text{textO}}{\underset{||}{C}} - OH$ & $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH$
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47. Which one of the following is not an amphoteric species? (1)
- A. HPO_4^{2-} B. H_2O
 C. NH_3 D. HNO_2
-
48. For the reaction below, which pair of species are both bases? $HCN + H_2O \rightleftharpoons H_3O^+ + CN^-$ (1)
- A. HCN & CN^- B. H_3O^+ & H_2O
 C. HCN & H_3O^+ D. H_2O & CN^-
-
49. What is the pH of a 0.01M weak base, B solution? $K_b = 1.6 \times 10^{-9}$ (1)
- A. 8.6 B. 5.6
 C. 4 D. 6
-
50. The concentration of hydroxide in a solution in which the concentration of hydroxide ion is 100 times hydronium ion at 25°C is ----- (1)
- A. 1.0×10^{-5} B. 1.0×10^{-6}
 C. 1.0×10^{-8} D. 1.0×10^{-9}
-
51. The pH of 0.1M solution of weak acid is 3.0. What is the value of the ionization constant for the acid? (1)
- A. 10^{-5} B. 10^{-6}
 C. 10^{-8} D. 10^{-10}
-
52. A buffer solution is a mixture of (1)
- A. a weak acid and a strong acid B. a strong acid and its conjugate base
 C. a weak acid and its conjugate base D. a strong acid and its conjugate acid

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61. What are the expected products of the electrolysis of an aqueous solution of CuSO_4 using an inert electrode? (1)
- A. Cu and H_2 at the cathode and dissolving of Cu at the anode
B. Cu at the cathode and O_2 at the anode only
C. Cu and H_2 at the cathode and only O_2 at the anode
D. Cu at the cathode, O_2 at the anode and H_2SO_4 solution
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62. In the electroplating of nickel from a solution containing Ni^{2+} ion, what will be the weight of the metal deposited on the cathode by a current of 8 A flowing for 500 minutes? (1)
- A. 73 g
B. 103 g
C. 117.4 g
D. 145 g
-
63. A 1M solution of $\text{Cu}(\text{NO}_3)_2$ is placed in a beaker with a strip of Cu metal. A 1M of SnSO_4 is placed in a second beaker with a strip of Sn metal. The two beakers are then connected by a salt bridge and the two metal electrodes are connected by wires to a voltmeter. Which of the following electrode serves as the anode and which electrode gains mass? Given that $E^\circ \text{Cu}^{2+}/\text{Cu} = 0.34\text{V}$ and $E^\circ \text{Sn}^{2+}/\text{Sn} = -0.14\text{V}$ (1)
- A. Sn, Sn electrode gains mass
B. Cu, Sn electrode gains mass
C. Sn, Cu electrode gains mass
D. Cu, Cu electrode gains mass
-
64. Which of the following is WRONG concerning a Galvanic cell? (1)
- A. It involves the deposition of one metal at one electrode when oxidations of another metal occur at the second electrode.
B. A spontaneous chemical reaction generates an electric current
C. A non-spontaneous chemical reaction generates electrical energy
D. It does not require an external source of electric current to function
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65. For the electrochemical cell represented by: $\text{Zn(s)}/\text{ZnSO}_4(\text{aq}) // \text{CuSO}_4(\text{aq})/\text{Cu(s)}$. (1)
The standard reduction potentials for Cu^{2+}/Cu and Zn^{2+}/Zn are + 0.34 and - 0.76V, respectively. What will be the cell reaction that occurs spontaneously and its cell voltage? (V)
- A. $\text{Cu} + \text{Zn}^{2+} \longrightarrow \text{Cu}^{2+} + \text{Zn}$ 1.10
B. $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$ 0.42
C. $\text{Cu}^{2+} + \text{Zn}^{2+} \longrightarrow \text{Zn} + \text{Cu}$ 0.42
D. $\text{Cu}^{2+} + \text{Zn} \longrightarrow \text{Cu} + \text{Zn}^{2+}$ 1.10
-
66. Which of the following statement describe Chemical industries? It is : (1)
- A. A process in which dissolved solute comes out of solution and form crystals
B. An area of chemistry concerned with the rate at which chemical reaction occur
C. A firm that involves the usage of raw material to form desired products by means of chemical reaction
D. An expression for the chemical composition of compound.
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67. Which of the following is a non-renewable resource: (1)
- A. soil
B. petroleum
C. water
D. plants
-
68. Which of the following is the most important ingredients used for production of DAP. (1)
- A. Ammonia and phosphoric acid
B. Phosphoric acid, urea and ammonia
C. Nitric acid and urea
D. Sulphuric acid, ammonia and urea

69. The hydrogen used in Haber process is made by the following reaction $CH_4 + H_2O \rightleftharpoons CO + 3H_2$, $\Delta H = +206\text{KJ}$. Which of the following set of condition will favor the formation of more H_2 Molecules? (1)
- A. Low pressure and high temperature
B. High pressure and high temperature
C. Low pressure and low temperature
D. High pressure and low temperature
70. Ammonia is produced in laboratory by heating ammonium chloride with _____. (1)
- A. Calcium nitrate
B. Calcium hydroxide
C. Calcium chloride
D. Calcium sulphate
71. Among the following compounds which one is a by-product in Solvay process during the production of Na_2CO_3 ? (1)
- A. $CaCl_2$
B. CO_2
C. NH_3
D. $CaCO_3$
72. Which of the following pairing is not correct? (1)
- A. Solvay process- $NaCl$
B. Ostwald process- HNO_3
C. Contact process- H_2SO_4
D. Haber process- NH_3
73. What is the purpose of adding ethanol during the preparation of soap? (1)
- A. To reduce the solubility of soap & separate it out
B. To dissolve the fat & make hydrolysis easily
C. To make soft and smooth
D. to make pleasant smell
74. The major composition of cement is (1)
- A. SiO_2
B. CaO
C. Fe_2O_3
D. Al_2O_3
75. The purpose of annealing during glass production is _____. (1)
- A. changing into required shapes
B. melting the raw material
C. testing the quality of glass
D. removing internal stress
76. The stage in which the shredded sugarcane travels on the conveyor belt through a series of heavy-duty rollers, which extract juice from the pulp in sugar manufacturing, is called _____. (1)
- A. Juicing
B. Cleansing and Grinding
C. Collecting the Harvest
D. Clarifying
77. In the process of sugar manufacturing which one of the following substance is used to bleach the brown syrup? (1)
- A. Sodium hypochlorite
B. Sulfur dioxide
C. Hydrogen peroxide
D. Chlorine
78. Of the following methods, which one is a traditional food preservation method? (1)
- A. Vacuum packing
B. Freeze drying
C. Salting
D. Freezing

— END OF EXAMINATION —